

Data Science & AI Bootcamp – Interview Question Bank

DEEP LEARNING

1. What is the difference between deep learning and machine learning?
2. Explain activation functions and their purposes.
3. What is backpropagation?
4. Explain different types of neural network layers.
5. What is gradient descent?
6. How do you handle overfitting in neural networks?
7. Explain batch normalization.
8. What is dropout and why is it used?
9. Explain the vanishing gradient problem.
10. What are loss functions?
11. How do you initialize weights in neural networks?
12. What is the role of bias in neural networks?
13. Explain epoch, batch, and iteration.
14. What is transfer learning?
15. How do you choose the number of hidden layers?
16. What is the difference between CNN and RNN?
17. Explain max pooling and average pooling.
18. What are optimizers in deep learning?
19. How do you handle imbalanced data in deep learning?
20. Explain early stopping.
21. Explain different CNN architectures (ResNet, VGG, etc.).
22. What are attention mechanisms?
23. Explain LSTM and GRU architectures.
24. What are autoencoders?
25. Explain different regularization techniques.
26. What is the transformer architecture?
27. How do you implement custom loss functions?
28. Explain various optimization algorithms.
29. What are embedding layers?
30. Explain self-attention mechanisms.
31. What are skip connections?
32. How do you handle sequential data?
33. Explain different normalization techniques.
34. What are adversarial attacks?
35. How do you implement transfer learning with custom architectures?
36. Identify the main components of a neural network.
37. What is a convolutional neural network (CNN)?

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- 38. Define a recurrent neural network (RNN).
- 39. What is a dropout layer in a neural network?
- 40. Discuss the role of optimizers in training neural networks.
- 41. What is the purpose of loss functions in deep learning?
- 42. Differentiate between L1 and L2 regularization techniques.
- 43. Explain how Long Short-Term Memory (LSTM) networks address limitations of traditional RNNs.
- 44. Identify common architectures used in deep learning for image classification tasks.
- 45. Discuss the role of hyperparameter tuning in deep learning models.
- 46. What is model ensembling, and why is it used?
- 47. What is the role of the encoder in a transformer model?